ETW3482 Predictive Modelling and Scoring Project Marking Rubric - Semester 2, 2026

Criteria	High Distinction (HD) (80-100%)	Distinction (D) (70-79%)	Credit (C) (60-69%)	Pass (P) (50-59%)	Fail (F) (<50%)
Predictive Model Exploration through Hyperparameter Investigation **(Section 2) 25%**	**21-25 points** Critically investigates meaningful hyperparameters across all required model families and accurately explains what the selected hyperparameters control. Justifies the settings explored, systematically evaluates how alternatives affect model behaviour and predictive performance, and integrates selective evidence across models. Synthesises the combined findings to develop well-supported and insightful conclusions about target-input relationships and the prediction problem. Exploration is purposeful, initiative-driven, and clearly extends beyond optimisation or procedural node execution.	**18-20 points** Analyses relevant hyperparameters across the required model families and demonstrates clear understanding of what they control. Justifies the main settings explored and accurately evaluates changes in model behaviour and performance. Synthesises evidence across models to draw supported conclusions about target-input relationships. Investigation is systematic and analytical, with only minor limitations in breadth, depth, integration, or critical insight.	**16-17 points** Applies generally appropriate hyperparameter changes and demonstrates a sound but uneven understanding of their roles. Compares model behaviour and performance with some relevant interpretation. Identifies plausible target-input relationships, but synthesis across models is limited, partially descriptive, or dependent on a narrow set of results. Justification of settings and analytical depth are inconsistent.	**13-15 points** Explores some relevant hyperparameters but the investigation is narrow, mechanical, or incomplete. Provides basic descriptions of model behaviour and performance with limited explanation of why settings were selected. Conclusions about target-input relationships are generic, weakly supported, or drawn from isolated results. Evidence and cross-model synthesis are limited.	**0-12 points** Provides inadequate or materially incorrect hyperparameter investigation. Demonstrates little understanding of what selected hyperparameters control, makes unsupported or inappropriate comparisons, or omits substantial required modelling. Model behaviour and performance are misinterpreted or insufficiently evidenced, and meaningful conclusions about target-input relationships are absent or unsupported.
Champion Model Identification **(Section 3) 10%**	**9-10 points** Critically selects and justifies one performance metric that is well matched to the prediction problem and suitable for comparing the candidate models. Applies the metric consistently, evaluates the comparative evidence accurately, and defends the champion-model selection with clear reasoning. Demonstrates strong understanding of why the selected evidence provides an appropriate basis for model choice rather than relying on automated software selection.	**8 points** Selects an appropriate performance metric and provides a clear, well-developed justification linked to the prediction problem. Compares candidate models correctly and identifies a well-supported champion model. Reasoning is analytical and evidence-based, with only minor limitations in depth or evaluation.	**7 points** Selects a generally appropriate metric and compares the candidate models mostly correctly. Provides some relevant justification, although the reasoning may be generic or only partly connected to the prediction problem. The champion-model choice is supported by the evidence but evaluation is limited.	**5-6 points** Uses a plausible metric but provides limited justification or applies the comparison inconsistently. Champion-model identification is basic, partly reliant on software output, or weakly defended. Understanding of why the metric is suitable for model comparison is only partially demonstrated.	**0-4 points** Selects an inappropriate or unsupported metric, compares models incorrectly or inconsistently, or fails to provide adequate model-comparison evidence. Champion-model identification is missing, unsupported, or materially inconsistent with the presented results.
Prediction Performance Insights **(Section 4.1) 10%**	**9-10 points** Selects a highly appropriate performance chart and critically justifies its relevance to the prediction objective. Interprets the champion model's predictive behaviour accurately and derives meaningful, evidence-based insights rather than describing chart features. Evaluates the practical usefulness and implications of the predictions, including relevant strengths, limitations, or trade-offs, and connects the findings clearly to the prediction objective.	**8 points** Selects and justifies an appropriate performance chart and interprets the main predictive patterns accurately. Derives clear prediction insights and connects them effectively to the objective and practical use of the model. Analysis is strong, with only minor limitations in depth or evaluative insight.	**7 points** Uses an appropriate chart and correctly interprets its main features. Provides some meaningful prediction insights and practical linkage, but discussion is partly descriptive or the justification and implications are underdeveloped.	**5-6 points** Presents a relevant chart but provides weak justification and mainly describes axes, values, or visible patterns. Prediction insights are limited, generic, or weakly connected to the prediction objective and practical use.	**0-4 points** Uses an inappropriate chart, materially misinterprets the chart, or provides little or no meaningful analysis. Prediction insights are absent, unsupported, or disconnected from the prediction objective.

ETW3482 Predictive Modelling and Scoring Project Marking Rubric - Semester 2, 2026

Criteria	High Distinction (HD) (80-100%)	Distinction (D) (70-79%)	Credit (C) (60-69%)	Pass (P) (50-59%)	Fail (F) (<50%)
Model Interpretability and Predictor Insights **(Section 4.2) 15%**	**13-15 points** Critically selects and applies interpretability tools that are well matched to the analytical questions. Accurately identifies and investigates the three most influential predictors and integrates interpretability evidence to explain their overall influence on champion-model predictions across the dataset. Evaluates relevant numerical patterns and/or categorical differences, synthesises the findings into meaningful target-input insights, and connects those insights convincingly to context and potential decision implications. Interpretation is precise, evidence-based, and analytically insightful.	**12 points** Selects appropriate interpretability tools and applies them correctly to identify and investigate the three most influential predictors. Explains the main predictor influences and patterns accurately and connects the evidence clearly to target-input insights, context, and decision implications. Analysis is well integrated, with only minor limitations in depth or synthesis.	**10-11 points** Uses generally appropriate and some inappropriate interpretability tools and identifies the main influential predictors. Interpretation of the principal patterns is mostly correct, but tool selection, integration of evidence, or explanation of predictor influence is uneven. Contextual meaning and decision implications are present but may be generic or only partly developed.	**8-9 points** Applies all basic interpretability tools and the analysis is general and limited, incomplete, or mainly descriptive. Identification or investigation of the influential predictors may be only partially supported. Relationships between interpretability evidence, target-input insights, context, and decision implications are weak or inconsistent.	**0-7 points** Fails to apply appropriate interpretability analysis to the champion model across the dataset, or materially misinterprets the outputs. The influential predictors are incorrectly identified, insufficiently investigated, or unsupported. Little meaningful connection is made between interpretability evidence, target-input relationships, context, and decision implications.
Model Deployment and Scoring **(Section 5) 10%**	**9-10 points** Accurately and completely deploys the selected champion model in SAS Model Manager, imports the scoring dataset, generates predictions, and produces a clean scored output retaining the required ID and relevant prediction fields. Selective evidence clearly verifies each key deployment and scoring stage. The submitted scored dataset is consistent with the deployed champion model and is readily usable by a downstream decision-making or operational process.	**8 points** Completes the deployment and scoring workflow accurately and produces a correct, usable scored dataset with appropriate supporting evidence. Minor presentation or documentation limitations do not affect the validity of the implementation or output.	**7 points** Completes most deployment and scoring requirements correctly. The scored output is generally usable, but there are minor inconsistencies, omissions, or weaknesses in the evidence, field retention, presentation, or alignment with the documented process.	**5-6 points** Attempts the required deployment and scoring workflow but contains notable implementation weaknesses or incomplete evidence. The scored output requires correction or clarification before it could be used reliably in a downstream process.	**0-4 points** Fails to complete a valid deployment and scoring workflow, deploys the wrong or unsupported model, or produces an incorrect, incomplete, or unusable scored dataset. Required evidence or key output fields are substantially missing.

ETW3482 Predictive Modelling and Scoring Project Marking Rubric - Semester 2, 2026

Criteria	High Distinction (HD) (80-100%)	Distinction (D) (70-79%)	Credit (C) (60-69%)	Pass (P) (50-59%)	Fail (F) (<50%)
Actionable Strategy and Implementation Considerations **(Section 6) 15%**	**13-15 points** Critically translates prediction outputs and interpretability insights into specific, evidence-based organisational strategies. Develops a clear and convincing Prediction -> Decision -> Action -> Expected organisational outcome pathway and explains how the solution would operate within a realistic workflow, including relevant users or functions. Evaluates implementation feasibility, organisational or operational challenges, consequences of incorrect predictions, and responsible-use considerations. Recommendations are coherent, practical, context-specific, and strongly supported by the analysis.	**12 points** Translates prediction and interpretability findings into clear, practical strategies with a well-developed connection between predictions, decisions, actions, and expected outcomes. Explains relevant workflow, users, implementation considerations, and risks. Recommendations are well aligned with the organisational context, with only minor limitations in depth, feasibility evaluation, or strategic integration.	**10-11 points** Proposes relevant actions and strategies with some connection to prediction and interpretability evidence. Describes a plausible organisational workflow and identifies important implementation considerations. However, parts of the strategy are generic, evidence linkage is uneven, or evaluation of feasibility, risks, and responsible use is underdeveloped.	**8-9 points** Provides basic or generic strategies with limited connection to the analytical findings. The prediction-to-action pathway, organisational workflow, implementation challenges, or consequences of incorrect predictions are only partially addressed. Recommendations are weakly justified or insufficiently operational.	**0-7 points** Provides absent, unrealistic, or largely unsupported strategies. Fails to translate predictive findings into meaningful organisational decisions and actions, or does not address implementation and operational considerations in a credible manner. Recommendations are disconnected from the analysis or the stated prediction problem.
Synthesis and Professional Communication **(Section 7 and Overall Report) 15%**	**13-15 points** Synthesises the major findings from model exploration, champion selection, prediction insights, interpretability, deployment, and operationalisation into a concise and evaluative conclusion that clearly identifies strengths, limitations, and practical considerations. Communicates Part A authentically for a Technical Manager and Part B effectively for a Business Decision Maker. Writing is precise, coherent, and selective. Tables, figures, and screenshots are purposefully integrated, clearly presented, and referred to in the discussion. The report is professionally structured and fully consistent with the stated format and word-limit requirements.	**12 points** Provides a strong synthesis of the major project findings and communicates clearly for the two intended audiences. Writing is coherent, technically appropriate, and professionally structured. Supporting visuals are well integrated and report requirements are followed, with only minor limitations in concision, synthesis, audience adaptation, or presentation.	**10-11 points** Summarises the main project findings accurately and communicates them understandably, but synthesis is uneven or partly repetitive. Audience adaptation is evident but inconsistent. Organisation, terminology, concision, or integration of tables, figures, and screenshots is generally adequate but not consistently effective.	**8-9 points** Provides a basic conclusion and report structure but mainly repeats earlier findings rather than synthesising them. Communication shows limited adaptation to the intended audiences. Writing, organisation, formatting, or use of supporting visuals contains notable weaknesses that reduce clarity and professionalism.	**0-7 points** Fails to provide a meaningful synthesis of the project or communicate the work effectively to the intended audiences. The report is substantially disorganised, repetitive, technically unclear, or non-compliant with key presentation requirements. Supporting evidence and visuals are missing, poorly integrated, or insufficient to support understanding.